

Rashik Intisar Siddiquee

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Career Summary

- Expertise in applying data science, statistical analysis, AI/ML solutions for decision optimization
- Research focuses on building machine learning models and translating results into actionable business insights
- Industry experience in building ETL pipelines and dashboards system using Python, SQL

Education

- **Ph.D. in Industrial Engineering**, North Carolina State University *Aug 2021 – Oct 2026*
Major: Operations Research — Minor: Statistics
- **MBA in Finance**, University of Dhaka *May 2021*
- **B.S. in Mechanical Engineering**, Bangladesh University of Engineering Tech *Oct 2018*

Technical Skills

Programming & Data:	Python, R, PostgreSQL, SQL Server, Pandas, NumPy, MATLAB
Machine Learning:	Supervised & Unsupervised Learning, Random Forest, Regression, Clustering
Deep Learning:	Neural Networks, PyTorch, TensorFlow
MLOps & Deployment:	CI/CD pipelines, Google Cloud, Virtual Machines, Git
Visualization:	Tableau, Power BI
Optimization:	Gurobi, Linear Programming, Mixed-Integer Optimization

Experience

Graduate Research Assistant, NC State University *Aug 2021 – Present*

- Designed and deployed end-to-end ML and optimization-driven systems for 2 local food bank operations
- Built scalable pipelines for integrating structured and unstructured data, including LLM-based workflows
- Developed 3 ML + optimization models for food allocation and agency decision support
- Deployed 2 AI/ML-based solutions and 3 decision analytics web dashboards

Business Intelligence Analyst, UCB Fintech Company Limited *Mar 2021 – Aug 2021*

- Designed 30+ ETL pipelines from ERP data using SQL, Python for 10 million customers
- Reduced system failure rate from 10% to 3% through data-driven monitoring and anomaly detection systems
- Developed Tableau dashboards to communicate insights and support decision-making across teams.
- Identified 3 revenue leakage issues and coordinated with tech and product team to solve them

Senior Executive - Digital Financial Services, CVC Finance Limited, Dhaka, Bangladesh *Aug 20 - Mar 21*

- Led the launch of tech-enabled health insurance products backed by an e-deposit product
- Conducted market analysis of products and developed strategies to optimize product offering and engagement

Projects

- **Generative AI Pipeline for Data Extraction and Standardization** *2024 – Present*
 - Built an AI-based data extraction system to transform unstructured operational data to structured data
 - Improved data quality and usability, enabling downstream analytics, monitored and reporting
 - Designed model evaluation and refinement workflows to improve extraction accuracy from 60% to 93%
 - Enabled scalable processing of 800+ records to support data-driven decision-making
- **Predictive Modeling of Collaboration Behavior** *2023 – Present*
 - Collected, performed feature engineering on 200 entity operations data
 - Developed logistic regression, ML models including Random Forest, Bagging, Boosting methods to predict the willingness to collaborate among agencies, reaching 80% accuracy
- **Optimization Model for Food Bank Resource Allocation** *2024 – Present*
 - Developed multi-objective MILP models with efficiency, equity, and supply constraints
 - Modeled service networks and accessibility using transit data and geospatial analysis
 - Identified underserved areas and provided food banks with actionable insights
 - Delivered decision-support tools to right-size the agency network, achieving **30%** increase in access
- **Geo spatial Tool to Locate Food Sources for Wake County** *2025-Present*
 - Developed a resource locator using data processed by Finetuned LLM, for 300+ agencies
 - Deployed and maintained as a part of the County-wide food security initiative

Publications

- Siddiquee, R, Ivy, J. S., Ozaltin, O. Y. (2027). FoodBridge: Connecting Communities through Automated Information Extraction from Food Bank Partner Websites, *Submitted for AAAI 2027*
- Siddiquee, R, Ivy, J. S., Ozaltin, O. Y. (2025). Modeling Willingness to Collaborate for Resource Sharing in Food Bank Networks. Socio-Economic Planning Sciences *Manuscript submitted for publication.*
- Chowdhury, N. R., Siddiquee, R. I., and Ivy, J. S. (2025). Influence of Norms in Alliance Characteristics of Humanitarian Food Agencies: Proceedings of the 2025 Winter Simulation Conference 10.1109/WSC68292.2025.11338890

Honors and Activities

President, INFORMS NCSU Chapter — Edward P. Fitts Fellowship (2023–24) — Fitts Fellowship (2021)